In neuroscience or cognitive psychology, a reasoning like: "the same transformation applies to different objects, but the only properties shared between these objects are P_1, ..., P_n. So the transformation must depend only on these common properties"? Use examples where n>1

Invariance Extraction: Property-Dependent Transformations

In the specific case you described, we are not talking about a generic geometric transformation (like rotation, which applies to any object regardless of what it is), but about a **conditioned transformation**. The brain notices that a certain transformation is activated *only and exclusively* when specific properties **P1, ..., Pn** are shared by objects that are otherwise completely different from one another.

In cognitive and computational neuroscience, this phenomenon is called **Semantic and Functional Invariance Extraction** (often linked to the concept of **abstract affordance** or **relational rule learning**). The brain does not just extract geometries; it *extracts the logical rule* that binds those specific properties to the transformation.

Example 1: The Principle of 'Elastic Deformation via Pressure' (Motor and Visual Neuroscience)

Imagine the brain observing different objects undergoing the same transformation: **they are compressed and then return to their original shape**. The objects are visually and heterogeneously completely diverse: a yellow kitchen sponge, a red party balloon, a feather pillow, and a rubber eraser.

- **Shared Properties (P1, P2):** The only features they have in common (which the brain extracts) are not color or shape, but P1 = *porous/hollow internal structure* and P2 = *flexible material*.
- **The Transformation:** Elastic compression.
- **What the brain does:** Through *invariance extraction*, the parietal cortex and the motor system deduce that the 'squeezing' transformation is not random, but **depends strictly and exclusively on properties P1 and P2**. The brain creates a predictive rule: if a new object X possesses P1 and P2, it will undergo the same transformation when subjected to pressure.

Example 2: The Transformation of 'Optical Refraction/Distortion' (Ventral Visual Stream)

The brain observes different objects that, when looked through, **induce a visual transformation of magnification or distortion of the background images**. The objects causing this transformation are a water droplet on a leaf, a piece of smooth sea glass on the beach, and an antique monocular lens.

- **Shared Properties (P1, P2):** The objects have entirely different shapes, origins, and sizes. The only properties in common are P1 = *material transparency* and P2 = *curved surface*.
- **The Transformation:** The deviation/distortion of light rays (refraction).

- **What the brain does:** Visual area V4 and the temporal areas responsible for material recognition extract P1 and P2 as the only constant variables. The brain understands that the optical transformation *mathematically depends* only on curvature and transparency, completely ignoring whether the object is water or glass, natural or artificial.

How Does the Brain Compute This Link?

At the neuronal level, this process requires an integration between the **ventral stream** (which identifies the properties P1, ..., Pn of the objects) and the **prefrontal and parietal areas** (which encode rules and actions).

1. **Co-occurrence Detection:** Neurons in the prefrontal cortex monitor the transformation. Every time the transformation occurs, a specific neural population activation pattern is triggered.
2. **Logical Intersection (Noise Cancellation):** The brain compares the neural activation vectors of object A, object B, and object C. The coordinates describing the differences (e.g., the yellow color of the sponge or the shape of the droplet) cancel each other out over time through synaptic inhibition.
3. **Isolating the Causal Core:** Only the synapses responding to P1, ..., Pn remain active. The brain concludes that these properties are the only *valid predictors* of the transformation.

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